Cornell Duffield Engineering

Systems Engineering, Cornell Duffield Engineering
313 Campus Rd • Ithaca, NY 14853 • winyap@cornell.edu

AI for Cities (AI4C): From Urban Data to Real-World Impact

Project Description: We develop urban AI and computational tools for understanding complex city systems. The project focuses on creating intelligent, scalable, and interactive models of cities that integrate heterogeneous urban data, including crowdsourced data from OpenStreetMap, street view and satellite imagery, and human mobility flows. We use data and AI to help researchers, planners, policymakers, and other stakeholders explore urban conditions, identify patterns, and evaluate potential interventions. Where is the most heat vulnerable area in New York City? How do building density and street layout affect the prevalence of non-communicable health conditions, such as social isolation, depression, and obesity? The broader goal is to build an urban intelligence ecosystem that makes advanced city modeling more accessible and supports better decisions in areas such as transportation, housing, climate resilience, and infrastructure.

Related topics:

- **Urban Digital Twin Platform**: Building scalable and efficient infrastructure to support visualization, analytics, and simulation for sustainable urban development
- **Spatial AI and Reasoning**: Application of machine learning and data analytics for high-resolution spatial understanding, including flow prediction, anomaly detection, and inference of urban patterns
- **Agentic Geospatial Intelligence**: Develop agentic tools, servers, and knowledge infrastructure to support frontier urban intelligence in cities
- **Policy & Deployment**: Translation of research into practice through partnerships with municipalities, pilot digital-twin deployments, and assessment of their impact on real-world planning decisions

Requirements: Everyone interested in cities is welcome. The project scope is diverse and designed to be accessible to students from a wide range of academic backgrounds and skill levels. Students may contribute across areas including AI engineering, geospatial modeling, 3D visualization, LLM development, scalable systems architecture, human-centered design, community and stakeholder engagement, policy and evaluation, ethnographic research, and formulating conceptual and theoretical models of how urban technologies are understood and used in practice.

The project provides a student-driven, hands-on research environment for MEng and undergraduate students interested in cities, artificial intelligence, spatial computing, digital-twin technologies, and social science. Students will work with real-world datasets and solve pressing urban challenges while collaborating across disciplines such as computer science, planning, design, policy, environmental science, and the social sciences. Depending on the selected role and individual experience, relevant technical skills may include Python, R, GIS, cloud platforms, React, Deck.gl, MapLibre, Next.js, D3.js, WebGL, PyTorch, GeoAI, spatial databases, LangChain, Hugging Face, OpenAI tools, MCP, RAG, and APIs.

Students with design and social-science backgrounds are particularly welcome to apply and may contribute through UI/UX research and design, user surveys, platform usability testing, interface and service design, participatory workshops, interviews, and focus-group discussions with community members and other stakeholders.

Cornell University Gao Lab is recruiting Summer Research Assistants for a unified program on AI for Cities (AI4C). Students will contribute to model development, urban data analytics, policy assessment, and stakeholder engagement across related projects. Depending on student interest, milestones may include developing your own digital app or AI project, presenting your project to stakeholders, creating your own code package or dataset, or publishing scholarly papers.

Project Leader: Dr. Winston Yap

Faculty PI: Prof. Oliver H. Gao
Google Scholar: Google Profile
Website: Personal Site
Email: winyap@cornell.edu
Office: Carpenter Hall B1C